

■ SOUTHERN LATITUDES

Three bright stars dominate the scene to the north, the highest of which is Altair in Aquila. Below it are Vega in Lyra, and Deneb in Cygnus. The three form a triangle known as the Summer Triangle due to its prominence in the northern-hemisphere summer sky. The constellation of Vulpecula, the fox, which is between Aquila and Cygnus, contains a relatively easy-to-spot planetary nebula, the Dumbbell Nebula (M27). It is seen through binoculars as a rounded patch of light.



A second well-known planetary nebula, the Ring Nebula (M57), is in Lyra and can be seen through a small telescope as a disk of light. The constellation Tucana, the toucan, is to the southeast. It contains the globular cluster 47 Tucanae and the irregular galaxy called the Small Magellanic Cloud.

Sagittarius and the rich star fields in the center of the galaxy are still high overhead. The galaxy's milky path flows through Scorpius to the southwest, then through Lupus and on to Crux and the southern horizon. Alpha (α) and Beta (β) Centauri are low on the southwestern horizon and mark the front legs of Centaurus, the centaur (half man, half horse). Alpha Centauri, also known as Rigil Kentaurus, is the third-brightest star in the sky; Beta Centauri, or Hadar, is 11th brightest. The two appear to be the same distance from us, but in reality they are vastly separated. Beta is about 525 light-years away, and Alpha only 4.3 light-years away. A small telescope will show that Alpha Centauri is a binary star consisting of two yellow and orange stars that orbit each other every 80 years. Alpha Centauri is sometimes described as the closest star after the Sun, but Proxima Centauri, a much fainter red dwarf star possibly associated with Alpha, is a little closer. Centaurus is one of two centaurs in the sky; the second is Sagittarius.

47 Tucanae

The globular cluster 47 Tucanae is the second-brightest in the sky. At 13,400 light-years away, it is also one of the closest to Earth. It contains several million stars and appears as a hazy star to the naked eye.



Cygnus Rift

The path of the Milky Way flows from Cepheus (top) to Scorpius (near the horizon), where it is noticeably brighter as we look toward the galaxy's center. The dark band of dust along its center line is the Cygnus Rift.

